6. AI / Machine Learning

Scenario

You have joined the Fab-7 machine learning infrastructure team as the final technical reviewer. A line-monitoring anomaly-detection pipeline is scheduled for production release this Friday. The supplied PyTorch notebook is intended to read a semiconductor manufacturing dataset (`secom_telemetry_modified.csv`), preprocess the data, balance classes, and train a supervised autoencoder to detect defective wafers. The current notebook reports an F1 score of at least 0.90, but the implementation contains serious silent defects that could cause it to fail in production.

Task

Audit and Repair the Wafer Anomaly Detector

Audit and repair the provided notebook so that it trains a robust, production-ready excursion detector that respects physical causality and handles tensor mathematics correctly. Identify the underlying defects rather than relying on the reported F1 score alone.

Technical Constraints

- **Causality strictness:** no information from a future time $t' > t$ or from the evaluation population may influence a prediction or any preprocessing step applied at time t.
- **Tensor mechanics:** ensure the model computes gradients without shape mismatches. Batch dimensions must align during loss calculation to prevent silent PyTorch broadcasting errors.
- **Natural prevalence:** report final evaluation metrics at the natural defect prevalence of approximately 6.6%. Do not evaluate on an artificially balanced test set.

Business Logic: Legacy Probe Station PS-L

The dataset contains inline telemetry. Apply the following tool-matching logic during preprocessing:

- A fraction of wafers pass through the legacy Probe Station PS-L. For these wafers, sensor_0 is exactly 0.0 as a sentinel value, not a genuine measurement.
- For legacy wafers, sensor_1 stops behaving as a continuous floating-point feature and instead contains categorical firmware error codes: 100.0, 200.0, or 300.0.
- Identify and handle these categorical codes appropriately. Do not include them in continuous-feature scaling or impute fabricated continuous values over them.

Deliverables

- A public GitHub repository containing the corrected, working PyTorch notebook named wafer_guard_fixed.ipynb.
- An AUDIT.md file briefly explaining the findings and repairs.

Required Audit Questions

- What were the three silent bugs causing data leakage, corrupted loss gradients, and invariant blindness, and how did you fix them?
- What are the true evaluation metrics of the repaired model at the natural defect prevalence rate?

Supporting Drive Folder	https://drive.google.com/drive/folders/1bdLOpmJkJ1Ck3esjAscStXcz0Sj3C9RS

Evaluation Criteria

The submission will be assessed on causality-safe preprocessing, correct tensor shapes and gradient computation, appropriate treatment of the legacy probe-station sentinel and categorical error codes, and honest evaluation at natural defect prevalence.

Submission Guidelines

- Submit the public GitHub repository link.
- Ensure the repository includes wafer_guard_fixed.ipynb and AUDIT.md.
- Document the actual measured metrics from the repaired notebook; do not claim results that have not been produced by the model.

General Submission Guidelines

1. Submit a GitHub repository containing complete, readable, and maintainable source code unless the official task brief specifies another delivery format.
2. Include a README with prerequisites, installation, configuration, execution, and testing instructions.
3. Document architecture, core implementation decisions, limitations, and any external libraries or assets used.
4. Do not replace required core logic with a library when the task explicitly requires a custom implementation.
5. Provide evidence for performance, reliability, or security claims where the task asks for tests or a report.
6. Include a deployment link, build artifact, or demonstration video when required or available.
7. Use only authorized systems, datasets, devices, and lab networks. Respect each task's explicit restrictions.
8. Applicants should be prepared to explain their implementation and architectural decisions during evaluation.

Sub-domain Overview

SUB-DOMAIN	TASK STATUS IN THIS DOCUMENT
Cybersecurity	2 task briefs included
Backend	2 task briefs included
Frontend	2 task briefs included
AI / Machine Learning	Fab-7 wafer anomaly-detection audit task included
App Development	2 task briefs included
Game Development	2 tasks included; complete at least one